100 Days of Linux - Day 068

Title: Organizing Scripts with Bash Functions

Today's Goal
Learn how to create, call, and reuse functions in Bash scripts.

Why This Matters
Functions help you avoid repeating code. Large automation scripts use functions to keep code organized, readable, and easier to maintain.

Concepts of the Day

Create a Function:

```
greet() {
  echo "Hello, Linux!"
}
```

Call the Function:

```
greet
```

Function with Argument:

```
welcome() {
  echo "Welcome, $1"
}
welcome Vishal
```

Beginner Lesson
A function is a reusable block of commands. Instead of writing the same code multiple times, write it once inside a function and call it whenever you need it.

Practice Questions

1. Create a script named functions.sh with a function named greet that prints 'Welcome to Linux'.

2. Call the greet function twice from the same script.

3. Create another function named show_date that displays the current date using the date command.

4. Create a welcome function that accepts a name as an argument and prints 'Hello, NAME!'. Test it with your own name.

5. Run the script and display your current working directory.

Hints

Define functions before calling them. Pass arguments just like you pass arguments to a script.

Mini Challenge

Create system_report.sh with three functions: show_user, show_directory, and show_date. Call all three to generate a simple system report.

Common Mistakes
Calling a function before it is defined, or adding spaces between the function name and parentheses.

Did You Know?
Many production Bash scripts contain dozens of functions to separate configuration, validation, logging, and deployment tasks.

Pro Tip
Keep each function focused on a single task. Small, reusable functions make debugging much easier.

Answer Key (Instructor Only)
Q1: greet(){ echo 'Welcome to Linux'; }
Q2: greet && greet
Q3: show_date(){ date; }
Q4: welcome(){ echo "Hello, $1!"; } && welcome Vishal
Q5: bash functions.sh && pwd

Homework
Create a script with at least four functions: welcome, show_user, show_date, and show_directory. Call each function in order.

Next Day Preview
Tomorrow you'll combine variables, user input, conditions, loops, and functions to build a complete interactive Bash application.